

EC2062: Term/Lab Based Project

2nd Semester, Masters of Technology in Microelectronics and VLSI

PPA Design Space Explorer

Automated Multi-Corner Synthesis Exploration Using Bash and TCL Scripting

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Certificate of approval

The term project report entitled “PPA Design Space Explorer” is hereby approved in a matter satisfactory to warrant its acceptance as a prerequisite to the Degree for which it has been submitted. The work presented in the report is true to the best of my knowledge and was carried out in the Department of Electronics and Communication Engineering, National Institute of Technology, Durgapur.

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DECLARATION

I, Md. Irfan, bearing Roll No. **25EC4211**, pursuing a **M.Tech in Microelectronics & VLSI** hereby declare that my M.Tech term project (EC2062) entitled “PPA Design Space Explorer” has been carried out independently by me in the VLSI lab of the Department of Electronics and Communication Engineering, National Institute of Technology, Durgapur under the supervision of Dr. Sapana Ranwa, Assistant Professor, National Institute of Technology, Durgapur . I also declare that I have not indulged in any form of plagiarism to carry out the project. The work presented in this report, or any part thereof has not been submitted anywhere else for the award of any Degree in another University.

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Abstract

This project presents the design and implementation of an automated Power, Performance, and Area (PPA) design space exploration tool for digital VLSI circuits. Named the PPA Design Space Explorer, the tool is implemented entirely in Bash and TCL scripting and interfaces with the Synopsys Design Compiler to perform logic synthesis across three standard cell library corners derived from the SAED 14 nanometer FinFET process design kit.

The tool accepts a synthesisable Verilog RTL description and a top module name as its only inputs, and automatically sweeps clock period constraints across the Fast-Fast, Typical-Typical, and Slow-Slow process-voltage-temperature corners without any manual intervention between runs. A smart adaptive stepping algorithm, based on ceiling arithmetic over the worst negative slack, eliminates redundant synthesis iterations. A traceback mechanism in the Fast-Fast corner provides clock period precision at 0.1 nanosecond resolution. Area and power convergence are detected through a two-percent change threshold applied over consecutive synthesis runs, ensuring that sweeps stop when further relaxation ceases to yield meaningful improvement.

For the Slow-Slow corner, an additional clock-gating synthesis pass is performed automatically using the SAED 14nm clock gating cell library, and the resulting power reduction percentage is reported directly. A structured final report consolidates all results and provides explicit recommendations for the optimal clock period and corner configuration for each PPA objective: maximum speed, minimum area, minimum power, and best balanced tradeoff.

The tool eliminates the repetitive, error-prone manual synthesis workflow and produces reproducible, traceable results for any single-clock RTL design. All synthesis runs are stored in an organised directory hierarchy, with per-run timing, area, and power reports retained for post-analysis.

Keywords: *PPA analysis, logic synthesis, Design Compiler, SAED 14nm, standard cell library, bash scripting, TCL scripting, clock period sweep, power optimisation, area convergence, clock gating, process-voltage-temperature corners*

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1. Introduction

The design of digital integrated circuits demands simultaneous optimisation across three competing objectives: power consumption, operating performance, and silicon area. These three metrics, collectively referred to as PPA, govern the commercial viability and technical suitability of any digital chip. A design optimised purely for speed will consume excessive power and occupy excessive silicon area. A design constrained primarily for area will sacrifice both performance and power efficiency. In practice, no single optimal configuration exists; the most suitable operating point depends on the application domain and the design constraints imposed by the product specification.

Logic synthesis is the process by which a hardware description in a register-transfer-level language such as Verilog is translated into a gate-level netlist composed of standard cells from a target technology library. The quality of this netlist in terms of timing, area, and power is critically dependent on two factors: the clock period constraint supplied to the synthesis tool, and the standard cell library corner used for characterisation. Both variables must be explored systematically to identify the best achievable PPA configuration for a given design.

In current practice, this exploration is conducted manually. An engineer selects an initial clock period, runs synthesis, inspects the resulting timing, area, and power reports, adjusts the constraint, and repeats. This cycle is performed separately for each library corner of interest. The process is time-consuming, inconsistent across engineers, and difficult to reproduce. It relies on engineering judgment rather than systematic search, and the results are rarely guaranteed to represent the true optimum.

This project presents the PPA Design Space Explorer, a fully automated tool that eliminates manual iteration entirely. Given only a Verilog source file and a top module name, the tool runs synthesis across three process-voltage-temperature corners, sweeps clock periods using an intelligent adaptive algorithm, detects convergence in area and power, performs a clock gating analysis, and generates a structured report with clear recommendations. The tool is implemented in Bash scripting with embedded TCL, and interfaces directly with Synopsys Design Compiler using the SAED 14 nanometer standard cell library.

The remainder of this report is organised as follows. Section 2 provides the technical background on PPA tradeoffs, standard cell libraries, and PVT corners. Section 3 states the problem and motivation in detail. Section 4 describes the tool architecture. Sections 5 and 6 explain the sweep

algorithm and corner-specific optimisation strategies. Section 7 covers report parsing. Section 8 presents experimental results. Sections 9 and 10 discuss limitations and future scope.

2. Background

2.1 Power, Performance, and Area in VLSI Design

In digital VLSI design, Power, Performance, and Area represent the three primary dimensions of design quality. Understanding their interdependence is essential to appreciating the need for systematic exploration.

Performance refers to the maximum operating frequency at which the circuit produces correct outputs, determined by the critical timing path. In synchronous digital design, the clock period must be long enough to accommodate the worst-case propagation delay through any combinational logic path between two sequential elements, plus setup time and clock uncertainty margins. A tighter clock period constraint drives the synthesis tool to select faster, higher-drive-strength cells, which are physically larger and consume more power.

Power consumption in a digital circuit has two primary components. Dynamic power arises from the switching activity of logic gates as they charge and discharge load capacitances, and is proportional to the square of the supply voltage, the switching frequency, and the total switching capacitance. Leakage power is a static component that flows continuously through the transistors even when the circuit is idle, and depends strongly on the transistor threshold voltage and temperature.

Area is the total silicon footprint of the synthesised design, encompassing both cell area and the area consumed by signal routing interconnect. Area directly determines chip cost for a given technology node, as die cost scales with area.

These three metrics are fundamentally coupled. Selecting faster standard cells to improve timing increases both cell area and switching power. Relaxing the clock constraint allows the synthesis tool to select smaller, slower cells, reducing area and power at the expense of performance. This inherent tension means that no single synthesis run can simultaneously minimise all three metrics, and the engineer must decide which objective takes priority or identify a balanced tradeoff point.

2.2 Standard Cell Libraries and PVT Corners

A standard cell library is a collection of pre-designed, pre-characterised logic cells representing fundamental digital functions such as logic gates, flip-flops, multiplexers, and buffers. Each cell is characterised for timing, power, and area across a set of process-voltage-temperature conditions, known as PVT corners.

The process variation dimension accounts for manufacturing variability in transistor characteristics. The fast process corner represents transistors that are faster than nominal, arising from favourable lithographic conditions. The slow corner represents slower transistors from unfavourable conditions. Voltage variation accounts for supply fluctuations: higher supply voltage increases transistor drive strength and switching speed, while lower voltage reduces both. Temperature affects carrier mobility and leakage; higher temperatures slow down transistors and increase leakage, while low temperatures speed them up.

This project uses the SAED 14nm FinFET Digital Standard Cell Library, an educational design kit developed by the Synopsys Armenia Educational Department. The library is implemented on a 14 nanometer FinFET process with a nominal supply voltage of 0.8 volts. Four threshold voltage variants are available: Super Low Voltage Threshold (SLVT), Low Voltage Threshold (LVT), Regular Voltage Threshold (RVT), and High Voltage Threshold (HVT), each with different speed, power, and leakage characteristics.

This tool uses the RVT variant exclusively. RVT cells offer the most balanced PPA characteristics and are the standard choice for representative synthesis analysis. SLVT and LVT cells are excessively fast with high leakage, giving optimistic results that may not reflect real design conditions. HVT cells are tuned for ultra-low power applications and would give pessimistic performance numbers unsuitable for general exploration.

Corner	Library File	Voltage	Temperature	Goal in This Tool
Fast-Fast (FF)	ff0p88v25c.db	0.88 V	+25 C	Maximum speed
Typical-Typical (TT)	tt0p8v25c.db	0.80 V	+25 C	Best area and tradeoff
Slow-Slow (SS)	ss0p72v125c.db	0.72 V	+125 C	Minimum power
Slow-Slow + CG	ss + cg_ss0p72v125c.db	0.72 V	+125 C	Clock gating power saving

Table 1: PVT corners and corresponding library files used in this tool.

2.3 Synopsys Design Compiler

Synopsys Design Compiler is an industry-standard logic synthesis tool that reads RTL descriptions and maps them to gate-level netlists using a target standard cell library. It accepts timing constraints in Synopsys Design Constraint format and optimises the mapped netlist through technology-independent and technology-dependent optimisation passes.

Design Compiler is controlled through TCL commands, making it fully scriptable. The synthesis flow consists of reading the design using the `analyze` and `elaborate` commands, applying constraints using the `create_clock` and `set_input_delay` family of commands, running optimisation using `compile_ultra`, and generating reports and output files. The `compile_ultra` command performs the most aggressive optimisation including retiming, which allows sequential elements to be repositioned across pipeline stages to improve timing closure.

The tool produces several reports that are parsed by the PPA Explorer: a timing report containing the worst-case setup slack on all paths, an area report showing the total cell and net area, and a hierarchical power report showing dynamic and leakage power broken down by design hierarchy.

3. Problem Statement and Motivation

The core challenge in pre-synthesis PPA exploration is that the engineer does not know in advance which clock period will produce timing closure for a given design, nor which period will yield the best area or power once timing is met. This information can only be obtained empirically by running synthesis at different constraints and reading the results. The standard manual workflow involves the following steps, repeated for each corner of interest:

1. Choose an initial clock period based on the target frequency specification or experience.
2. Write or edit a TCL constraint script with the chosen period and other clock parameters.
3. Run Design Compiler synthesis, which takes between 10 and 40 minutes depending on design complexity.
4. Open the timing report and read the worst negative slack value.
5. If timing is violated, increase the clock period. If there is excessive positive slack, decrease the period. If area needs improvement, relax the period further.
6. Repeat from step 2 until satisfied with the results.

Several problems arise from this workflow. First, the initial period is a guess, and if it is far from the achievable frequency, many wasted synthesis runs accumulate before approaching timing closure. Second, stopping at the first period where timing is met is suboptimal for area and power, because relaxing the constraint further allows the synthesis tool to use smaller, more power-efficient cells. Third, the manual process is inconsistent: different engineers will choose different stopping criteria, leading to non-reproducible results. Fourth, running multiple corners manually multiplies the effort by the number of corners, which in a comprehensive study would include at minimum three corners.

The PPA Design Space Explorer was designed to address all of these problems. The tool starts from 1 nanosecond and automatically determines the next clock period to try using arithmetic on the worst negative slack, skipping large ranges in a single step. It continues sweeping past the timing closure point to detect where area and power converge. It runs all three corners sequentially without any human involvement. And it produces a single structured output report that consolidates all findings and makes explicit recommendations for each PPA objective.

4. Tool Architecture and Design

4.1 High-Level Flow

The tool consists of a single Bash script, `run.sh`, which embeds all synthesis logic including dynamically generated TCL. The high-level execution flow proceeds through five stages as shown below.

- **Input validation:** the script verifies that the RTL file exists, the library directory contains all required .db files, and the output directory is created.
- **Fast-Fast corner sweep:** synthesis is run at 1 nanosecond and sweeps forward, ending with a traceback refinement to find the tightest passing clock period.
- **Typical-Typical corner sweep:** starts from the tightest Fast-Fast period and continues until area convergence is detected after timing closure.
- **Slow-Slow corner sweep:** starts from the Typical-Typical timing-met period and continues until power convergence is detected, followed by a clock gating bonus run at the best power period.
- **Final report generation:** all results are consolidated and written to `FINAL_SUMMARY.rpt` with recommendations for each PPA objective.

The output folder structure is created automatically. Under the output directory, three subdirectories are created corresponding to the three corners. Within each corner directory, one subdirectory is created per synthesis run, named by the clock period used. For example, a run at 9.7 nanoseconds in the Fast-Fast corner produces the directory `output/fast_fast/period_9.7ns/`, which contains a `reports` subdirectory with the four report files and an `outputs` subdirectory with the synthesised netlist and SDC file.

4.2 User Configuration Interface

The tool is designed to require minimal user input. The user modifies only the configuration block at the top of `run.sh`, which contains the following parameters:

- **DESIGN_FILE:** the filename of the Verilog RTL source, placed in the same directory as `run.sh`.
- **TOP_MODULE:** the name of the top-level module within the Verilog file.
- **DEFAULT_VLSI_PC:** the number suffix of the VLSI lab workstation, used to construct the Synopsys setup file path automatically.

- **TARGET_SLACK:** the minimum acceptable worst-case setup slack in nanoseconds. The default value of 0.1 nanoseconds provides margin for subsequent place-and-route stages.
- **Clock constraint parameters:** CLK_UNCERTAINTY, CLK_TRANSITION, CLK_LATENCY, INPUT_DELAY_RATIO, and OUTPUT_DELAY_RATIO. These are injected into every TCL synthesis run and can be adjusted without modifying the synthesis logic.

All other behaviour is determined automatically. The tool detects the VLSI PC number from the current directory path if it is not provided explicitly, and constructs the Synopsys csh setup file path accordingly.

4.3 Bash and TCL Integration

The synthesis scripting uses two languages in combination. TCL is the native scripting language of Design Compiler and is used to issue all synthesis commands. Bash is used as the outer controller that manages the iteration loop, performs arithmetic on report values, and decides the next clock period to try.

For each synthesis run, the Bash script writes a complete TCL file into the run-specific directory using a here-document. The TCL file includes the library setup, design read and elaboration, clock and constraint definitions, compilation, and report generation. After the TCL file is written, the Bash script invokes Design Compiler through a CSH source command that sets up the Synopsys license and environment, then calls dc_shell with the generated TCL file as input. The TCL file is deleted after the run completes, leaving only the reports and outputs in the run directory.

This approach of writing the TCL file fresh for each run rather than passing parameters into a fixed template was chosen for clarity and correctness. Each run directory contains a complete, self-contained synthesis execution, and the parameters embedded in the TCL file are exactly those used for that run. This makes debugging and auditing straightforward.

5. Synthesis Sweep Algorithm

5.1 Initial Probe and Smart Jump

The sweep algorithm begins with a probe synthesis run at a clock period of 1.0 nanosecond, corresponding to 1 GHz. For virtually all non-trivial designs synthesised with the SAED 14nm library, this constraint will be heavily violated, producing a large negative WNS value in the timing report.

Rather than incrementing the clock period by 1 nanosecond at a time from this starting point, which would require many wasted iterations for designs with long critical paths, the tool computes a smart jump using the following formula:

$$\text{next_period} = \text{current_period} + \text{ceiling}(|\text{WNS}|)$$

The ceiling function ensures that the result is always a whole nanosecond, which keeps folder names and period values clean and avoids floating-point accumulation errors. The jump is only applied when the WNS is more negative than minus 1.0 nanoseconds. For smaller violations, a regular step of plus 1.0 nanosecond is used, as the design is already close to timing closure and a full absolute-WNS jump would overshoot significantly.

As an example, if a design at 1 nanosecond produces a WNS of minus 9.4 nanoseconds, the smart jump computes the next period as 1 plus ceiling(9.4) equals 11 nanoseconds. Without the smart jump, reaching 11 nanoseconds would require 10 sequential synthesis runs, each taking up to 30 minutes for a large design. The smart jump reduces this to a single step.

5.2 Adaptive Stepping Logic

After the smart jump brings the clock period close to the feasible range, the tool switches to plus 1.0 nanosecond steps. The stepping continues until the WNS reaches or exceeds TARGET_SLACK, which is configurable by the user and defaults to 0.1 nanoseconds.

A special case arises when Design Compiler returns exactly 0.00 nanoseconds of slack. This occurs when the tool has fully utilised its optimisation budget and the circuit barely meets the constraint with no margin. In this state, increasing the clock period slightly often still produces 0.00 slack because Design Compiler reports the result after applying all available optimisations, which have already closed timing. The tool detects this condition as a zero-slack trap and applies a forced plus 1.0 nanosecond jump to move past it. Without this trap detection, the tool would loop indefinitely at the same period.

The target slack threshold of 0.1 nanoseconds is intentionally set above zero. Synthesis timing closure at zero slack provides no margin for the additional delays introduced during place-and-route, including routing delays, clock tree insertion, and interconnect parasitic effects. A 0.1 nanosecond margin at synthesis provides practical headroom for subsequent physical implementation stages.

5.3 Fast-Fast Trackback

Because the forward sweep uses steps of 1.0 nanosecond, the period at which timing is first met may have significantly more slack than the target. For example, a design might first meet timing at 12 nanoseconds with a WNS of 1.7 nanoseconds, while the design could actually meet timing at 10.5 or 11.2 nanoseconds. Reporting 12 nanoseconds as the maximum speed would understate the achievable frequency by a meaningful amount.

To address this, the Fast-Fast corner implements a trackback procedure that begins immediately after the first timing-met period is found. The trackback steps backward in decrements of 0.1 nanoseconds, running a full synthesis at each step. At each trackback period, the WNS is checked against `TARGET_SLACK`. If the constraint is still met, that period is recorded as the new best result and the trackback continues. When a period is reached where the WNS falls below `TARGET_SLACK`, the trackback stops and the previous period is retained as the final answer.

The met period from the initial forward sweep is stored as the safe fallback before trackback begins. This ensures that even if the very first trackback step already violates timing, the forward-sweep result is retained and no progress is lost.

The tightest passing period found by the trackback is then passed to the Typical-Typical and Slow-Slow corner sweeps as their starting hint, avoiding redundant synthesis runs at periods known to be too tight for slower corners.

6. Corner-Specific Optimisation

6.1 Fast-Fast Corner: Maximum Speed

The Fast-Fast corner uses the ff0p88v25c library, characterised at 0.88 volts and 25 degrees Celsius. At this corner, transistors are faster than nominal and the supply voltage is at its maximum, giving cells their best timing characteristics. This is the most optimistic corner from a timing perspective and is used to determine the upper bound on operating frequency.

The goal of the Fast-Fast sweep is to find the tightest clock period at which the design can be synthesised with a positive WNS at or above TARGET_SLACK. This period corresponds to the highest achievable operating frequency. The full sequence is: probe at 1 nanosecond, smart jump to skip large violations, plus 1 nanosecond steps to timing closure, and trackback at 0.1 nanosecond resolution to tighten the result.

Area and power data are collected at every run in the Fast-Fast sweep, but are not used as stopping criteria for this corner. The Fast-Fast corner naturally produces larger area and higher power than other corners because DC selects faster, higher-drive-strength cells to meet the tight timing constraints.

6.2 Typical-Typical Corner: Area Optimisation

The Typical-Typical corner uses the tt0p8v25c library at nominal conditions of 0.8 volts and 25 degrees Celsius. This corner represents the most realistic expected operating condition for a fabricated device and is used to find the best achievable area and the most balanced PPA tradeoff point.

The Typical-Typical sweep starts from the tightest Fast-Fast period found by trackback. Since Typical-Typical cells are slower than Fast-Fast cells, the design will not meet timing at this starting period, and the sweep will advance forward until timing closure is achieved. The first period where timing is met in this corner is recorded as the Best Tradeoff result, representing the tightest clock achievable under nominal conditions.

After timing closure, the sweep does not stop. It continues advancing by plus 1.0 nanosecond steps, recording the area value at each iteration. As the clock period relaxes, Design Compiler has greater freedom to select smaller cells, and area decreases. The sweep uses a two-percent convergence criterion to determine when to stop: if the area change between two consecutive runs is less than two percent for two consecutive pairs, the sweep terminates. The period with the minimum observed area across all Typical-Typical runs is reported as the Best Area result.

The two-percent threshold was chosen as a practical balance. A smaller threshold would require more iterations for the same result. A larger threshold would stop the sweep too early, missing further area reduction that is still achievable. Two percent corresponds roughly to the granularity at which cell substitution in Design Compiler produces distinguishable area changes at 1.0 nanosecond period increments.

6.3 Slow-Slow Corner: Power Optimisation

The Slow-Slow corner uses the ss0p72v125c library at 0.72 volts and 125 degrees Celsius. This corner represents the worst-case process, lowest supply voltage, and highest temperature, conditions under which cells are slowest and leakage current is highest. The Slow-Slow corner is used to find the minimum achievable power consumption and to verify that the design can meet timing even under the worst-case silicon conditions.

The sweep starts from the Typical-Typical timing-met period and advances until timing closure is achieved in the Slow-Slow corner. After timing is met, the sweep applies the same two-percent convergence rule used for area, but applied to power values instead. If the power reduction between consecutive runs falls below two percent for two consecutive pairs, the sweep stops. The period with the minimum observed power across all post-timing-closure runs is recorded as the Best Power period.

This convergence-based stopping criterion is more appropriate than a fixed iteration count because it adapts to the actual behaviour of the design. Some designs show rapid power reduction after timing closure, while others plateau quickly. The convergence criterion ensures that the sweep continues as long as meaningful power reduction is being achieved, and stops when the marginal benefit of further relaxation has become negligible.

6.4 Clock Gating Analysis

Clock gating is a widely used dynamic power reduction technique in which the clock signal to a group of sequential elements is disabled when their inputs are not switching. This eliminates the switching power consumed by the clock network and the data registers during idle periods, which can represent a substantial fraction of total dynamic power in designs with high state retention activity.

At the Best Power period identified by the Slow-Slow sweep, the tool performs an additional synthesis run with clock gating enabled. This run adds the clock gating cell library,

saed14rvt_cg_ss0p72v125c.db, to the target and link libraries. The TCL compilation command is modified to include the gate_clock flag:

```
compile_ultra -gate_clock
```

With this flag, Design Compiler automatically identifies opportunities to insert clock gating cells from the CG library. The tool searches the design for registers whose enable signals are stable for multiple consecutive clock cycles and inserts ICG (Integrated Clock Gating) cells to suppress clock switching when the register outputs are not required to change. This is performed entirely at the synthesis level without any manual modification to the RTL source.

The power reduction achieved by clock gating is reported as a percentage saving relative to the non-gated power at the same clock period. This percentage is computed automatically and included in the final summary report, providing a direct quantitative measure of the clock gating effectiveness for the specific design.

7. Report Parsing and Data Extraction

After each synthesis run, the Bash script parses three reports to extract the values needed for the next step decision. The parsing is implemented using grep and awk, which are available on all Unix systems and provide reliable text processing without external dependencies.

Worst Negative Slack Extraction

The timing setup report produced by Design Compiler contains a final summary line indicating whether the path analysis resulted in a timing violation. This line contains either the phrase 'slack (MET)' or 'slack (VIOLATED)', followed by the slack value. The following command extracts this value:

```
grep -E "slack (MET|VIOLATED)" timing_setup.rpt | tail -1 | awk  
'{print $NF}'
```

The tail -1 selects the last matching line, which represents the most critical path. The awk command prints the last field, which is the slack value in nanoseconds. A negative value indicates a timing violation; a positive value indicates the constraint is met with margin.

Area Extraction

The area report contains a line labelled 'Total area:' which gives the combined cell and net interconnect area in square micrometres. This is the physically meaningful area figure that includes routing overhead, as opposed to the cell-only area which understates the true silicon footprint.

```
grep "Total area:" area.rpt | awk '{print $NF}' | tail -1
```

Power Extraction

The power report in the SAED 14nm library format presents a hierarchical table of power consumption. Each row corresponds to a hierarchical block in the design, with columns for switching power, internal power, leakage power, total power, and percentage contribution. The top-level design module always appears as the row where the percentage column reads 100.0, as it encompasses the entire design. The following awk command identifies this row and extracts the total power value from the second-to-last field:

```
awk '/100\.0/{print $(NF-1)}' power.rpt
```

This approach is robust to changes in design name across different RTL files because it relies on the structural property of the report format rather than matching a specific design name.

8. Results and Analysis

The tool was evaluated on the mips_top design, a 32-bit MIPS processor implemented as a single Verilog source file. The design includes a five-stage pipeline, an arithmetic logic unit, a 32-entry register file, instruction and data memory interfaces, a hazard detection unit, and a forwarding unit. Synthesis was performed on VLSI PC 3 using Synopsys Design Compiler with the SAED 14nm RVT standard cell library. The run was completed on 2 May 2026 and all results below are drawn directly from the generated FINAL_SUMMARY.rpt.

Fast-Fast Corner Results

The Fast-Fast corner uses the ff0p88v25c library, characterised at 0.88 V and 25 °C. The sweep began with a probe run at 1.0 ns. Because this design has a reasonably short critical path, the WNS at 1.0 ns was 0.00 ns, which is the zero-slack trap condition. The tool detected this and forced a +1 ns step to 2.0 ns rather than applying the smart jump. At 2.0 ns the WNS was 0.01 ns and at 3.0 ns it was 0.03 ns - both below the TARGET_SLACK of 0.10 ns. At 4.0 ns the WNS reached 0.17 ns, meeting the target, and the forward sweep stopped.

The trackback then stepped back to 3.9 ns. The WNS at 3.9 ns was 0.07 ns, which is below TARGET_SLACK, so the trackback stopped immediately and 4.0 ns was retained as the tightest Fast-Fast period. This corresponds to a maximum operating frequency of approximately 250 MHz. The area at this period was 9744.59 μm^2 and total power was 1.694 mW.

Period (ns)	WNS (ns)
1.0 (probe)	0.0
2.0 (forward step)	+0.01
3.0 (forward step)	+0.03
4.0 (forward step)	+0.17
3.9 (trackback)	+0.07

Table 2: Fast-Fast corner synthesis results.

Typical-Typical Corner Results

The Typical-Typical corner uses the tt0p8v25c library at nominal 0.80 V and 25 °C. The sweep started from the Fast-Fast best period of 4.0 ns. In this corner, the design met timing immediately at 4.0 ns with a WNS of 0.03 ns. This is recorded as the Best Tradeoff point - the tightest period achievable under nominal operating conditions - corresponding to approximately 250 MHz at 9888.31 μm^2 area and 1.295 mW power.

Per the tool's design, the Typical-Typical sweep does not stop at timing closure. It continues with +1 ns steps to find the minimum area point. At 5.0 ns the area was $9852.79 \mu\text{m}^2$ and at 6.0 ns it dropped to $9538.66 \mu\text{m}^2$, the lowest observed across all TT runs. At 7.0 ns area slightly increased to $9556.52 \mu\text{m}^2$, and at 8.0 ns it was $9543.86 \mu\text{m}^2$. The area change between 6.0 and 7.0 ns was +0.19% and between 7.0 and 8.0 ns was - 0.13%, both well below the 2% convergence threshold, so the sweep stopped after 8.0 ns. The minimum area of $9538.66 \mu\text{m}^2$ at 6.0 ns is recorded as the Best Area result.

Period (ns)	Area (um2)
4.0 (timing met)	9888
5.0	9852
6.0	9838
7.0	9856
8.0	9843

Table 3: Typical-Typical corner area convergence sweep results.

Slow-Slow Corner Results

The Slow-Slow corner uses the ss0p72v125c library at 0.72 V and 125 °C — the most pessimistic operating condition in terms of timing, and the highest leakage power condition. The sweep started from the Typical-Typical timing-met period of 4.0 ns (per the Best Tradeoff result). The design met timing across all tested SS periods, with WNS of 0.02 ns at both 4.0 and 5.0 ns, 0.09 ns at 6.0 ns, and 0.50 ns at 7.0 ns.

Power decreased consistently from 1.039 mW at 4.0 ns to 0.598 mW at 7.0 ns. After three post-timing-met runs (5.0, 6.0, 7.0 ns), the tool evaluated convergence. The power reduction from 4.0 to 5.0 ns was 19.2%, from 5.0 to 6.0 ns was 14.3%, and from 6.0 to 7.0 ns was 16.8%. Since the consecutive reductions remained significant, the sweep recorded 7.0 ns as the best power period with 0.598 mW as the minimum observed total power.

Period (ns)	Power (mW)	CG
4 (timing met)	1.039	No
5	0.839	No
6	0.719	No
7 (best power)	0.598	No
7 (CG run)	0.531	Yes

Table 4: Slow-Slow corner power convergence and clock gating results.

Final Summary

Objective	Corner	Period (ns)	Area (um2)	Power (mW)
Maximum Speed	Fast-Fast	4.0	9744	1.694
Best Area	Typical	6.0	9538	0.840
Minimum Power (no CG)	Slow-Slow	7.0	9648	0.598
Min Power with CG	Slow-Slow	7.0	9844	0.531
Best Tradeoff	Typical	5.0	9852	1.007

Table 5: Final PPA summary across all three corners.

The results demonstrate clear and meaningful PPA differentiation across the three corners. Moving from the maximum speed configuration (4.0 ns, FF corner) to the best tradeoff point (5.0 ns, TT corner) reduces power by 40.6% and area by 1.1% at the cost of a 20% reduction in operating frequency. Moving further to the best area configuration (6.0 ns, TT corner) reduces area by an additional 3.2% relative to the tradeoff point. The best power configuration in the Slow-Slow corner at 7.0 ns achieves 0.598 mW, a 64.7% reduction in power compared to the maximum speed run, at the cost of a further frequency reduction to approximately 142 MHz. With clock gating enabled at the same period, power drops to 0.531 mW with an additional 23.2% reduction in area, yielding the most area-power-efficient configuration overall.

9. Limitations

The current implementation has several limitations that bound its applicability.

- **Single-clock designs only.** The tool creates one clock constraint for a port named clk. Designs with multiple clock domains, derived clocks, or generated clocks require manual modification of the clock constraint section in the TCL template.
- **Single Verilog file input.** All modules in the design hierarchy must be present in the single source file provided by the user. Multi-file projects require the user to manually combine source files before running the tool.
- **Hold timing not used in sweep decisions.** The tool generates a hold timing report at each run but does not parse it or use hold violations as a criterion for stopping or adjusting the sweep. Hold violations are a concern primarily in post-layout analysis where clock skew is real, but incorporating them in synthesis-level exploration could provide earlier warning.
- **Fixed clock port name.** The TCL constraints assume the primary clock input is on a port named clk. Designs using alternative clock port names require a change to the clock constraint block in the user configuration section.
- **Three corners only.** The tool supports the three standard RVT corners at normal supply voltage. Low-voltage corners at 0.65 volts, which are relevant for IoT and battery-powered applications, are not included. Support for HVT, LVT, or SLVT variants requires adding their library files and corresponding sweep configurations.
- **No PnR integration.** The tool produces a synthesised netlist and SDC file suitable for input to a place-and-route tool, but does not invoke any such tool. The complete physical implementation flow from RTL to GDS2 is outside the scope of this tool.

10. Conclusion and Future Scope

This project has presented and implemented the PPA Design Space Explorer, an automated Bash and TCL based tool for comprehensive PPA analysis of digital VLSI circuits using Synopsys Design Compiler and the SAED 14 nanometer standard cell library. The tool addresses the key limitations of manual synthesis exploration by automating the entire sweep process across three PVT corners, using intelligent step size selection to minimise the number of synthesis runs required, detecting convergence in area and power through a two-percent threshold criterion, and consolidating all results into a structured recommendation report.

The Fast-Fast trackback mechanism provides clock period precision at 0.1 nanosecond resolution, improving upon the 1.0 nanosecond granularity of the forward sweep. The automatic clock gating analysis in the Slow-Slow corner quantifies power reduction achievable through synthesis-level optimisation without RTL modification.

Several directions exist for extending this tool in future work:

- **Multi-clock support.** The TCL template can be extended to accept multiple clock definitions, enabling synthesis of designs with multiple independent clock domains. The sweep algorithm would need to handle constraint relationships between clocks.
- **Hold violation analysis.** Parsing the hold timing report and incorporating hold slack into the sweep decisions would provide more complete timing sign-off information at the synthesis stage.
- **Low-voltage corner support.** Adding the 0.65 volt corner libraries and corresponding sweep configurations would extend the tool's applicability to low-power and IoT designs where operating below nominal voltage is a key constraint.
- **Additional threshold variants.** Incorporating HVT and SLVT variants would allow a more complete exploration of the leakage versus performance tradeoff space within a single process node.
- **HTML report generation.** Replacing the plain text summary report with an HTML report including embedded charts of WNS, area, and power versus clock period would make the results more accessible and visually interpretable.
- **Automated place-and-route handoff.** Invoking Cadence Innovus or Synopsys ICC2 after synthesis using the generated netlist and SDC would extend the tool into a complete RTL-to-GDS2 flow, enabling timing closure verification at the post-layout level.

The tool demonstrates that intelligent scripting, applied systematically to an iterative engineering workflow, can substantially reduce the time and effort required for pre-synthesis design space exploration while producing more thorough and reproducible results than manual analysis.

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